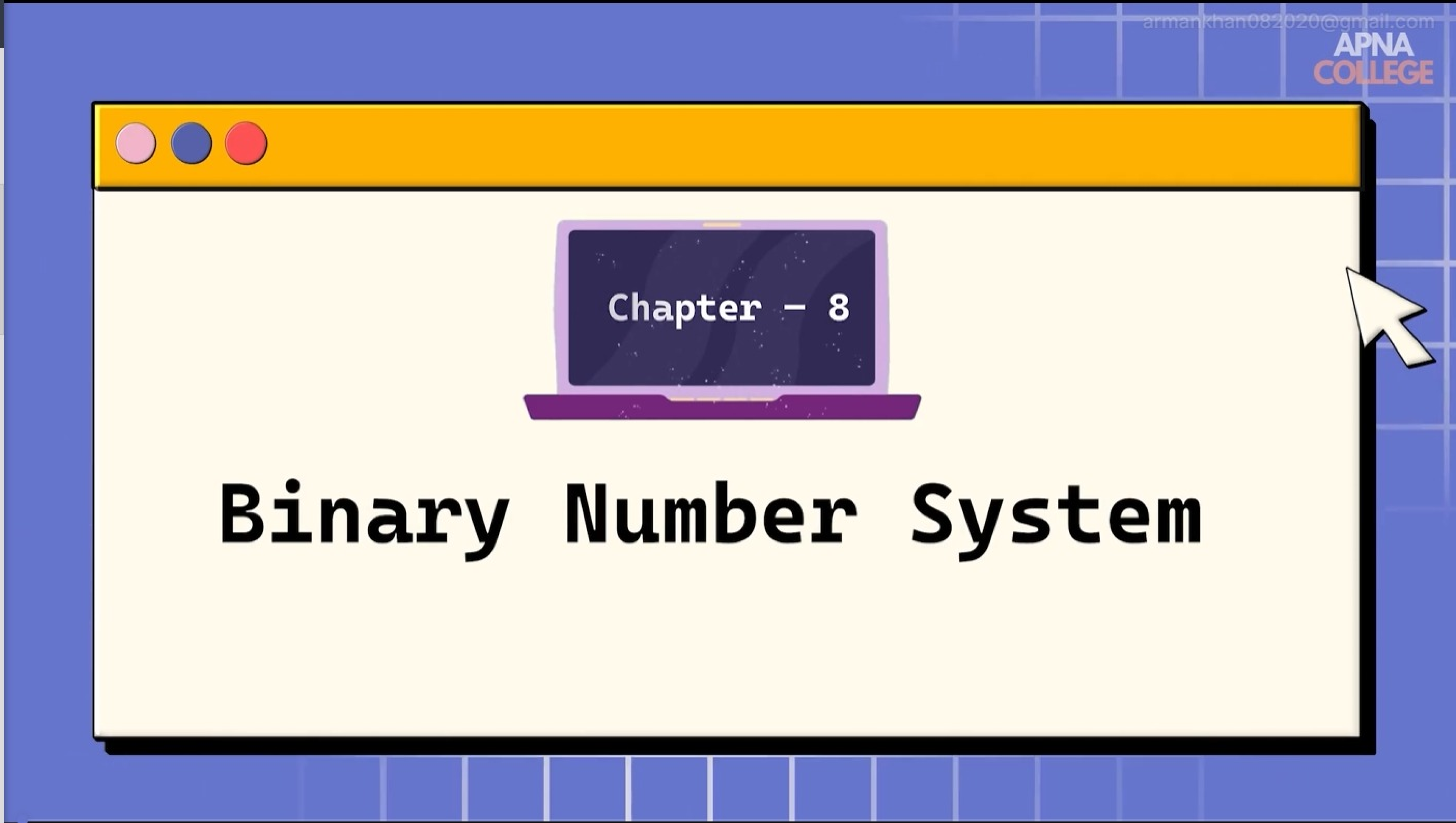
Sigma 9.0 (C++ DSA)

12 %



2

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Binary Number System

math → 0 to 9 ⇒ 10 digits

↳ Decimal

Binary Number System

int x :

$$100101 = 32 + 0 + 0 + 4 + 1 = 37$$

$$532 = 5 \times 100 + 3 \times 10 + 2 \times 1$$

$$5 \times 10^2 + 3 \times 10^1 + 2 \times 10^0$$

Binary Number System

18

Binary to Decimal

Convert 10010 to decimal representation.

$$\begin{array}{ccccc} 1 & 0 & 0 & 1 & 0 \\ 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$16 + 0 + 0 + 2 + 0$$

$$= \underline{18}$$

Binary Number System

Binary to Decimal

Convert 10010 to decimal representation.

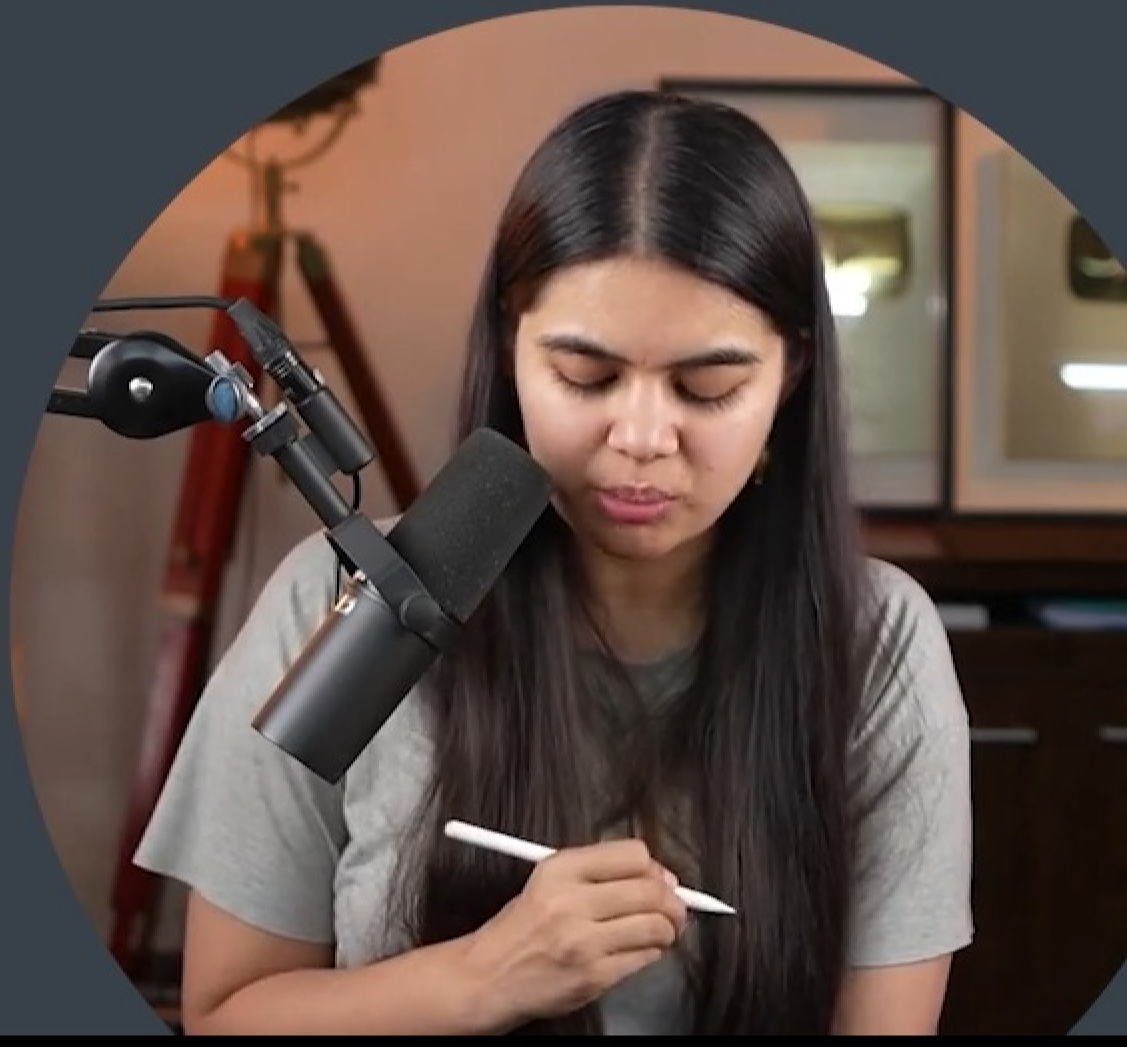
$$(18)_{10} = (10010)_2$$

0 0 0 1 0 0 1 0

$$\begin{array}{ccccc} 1 & 0 & 0 & 1 & 0 \\ 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$16 + 0 + 0 + 2 + 0$$

$$= \underline{18}$$



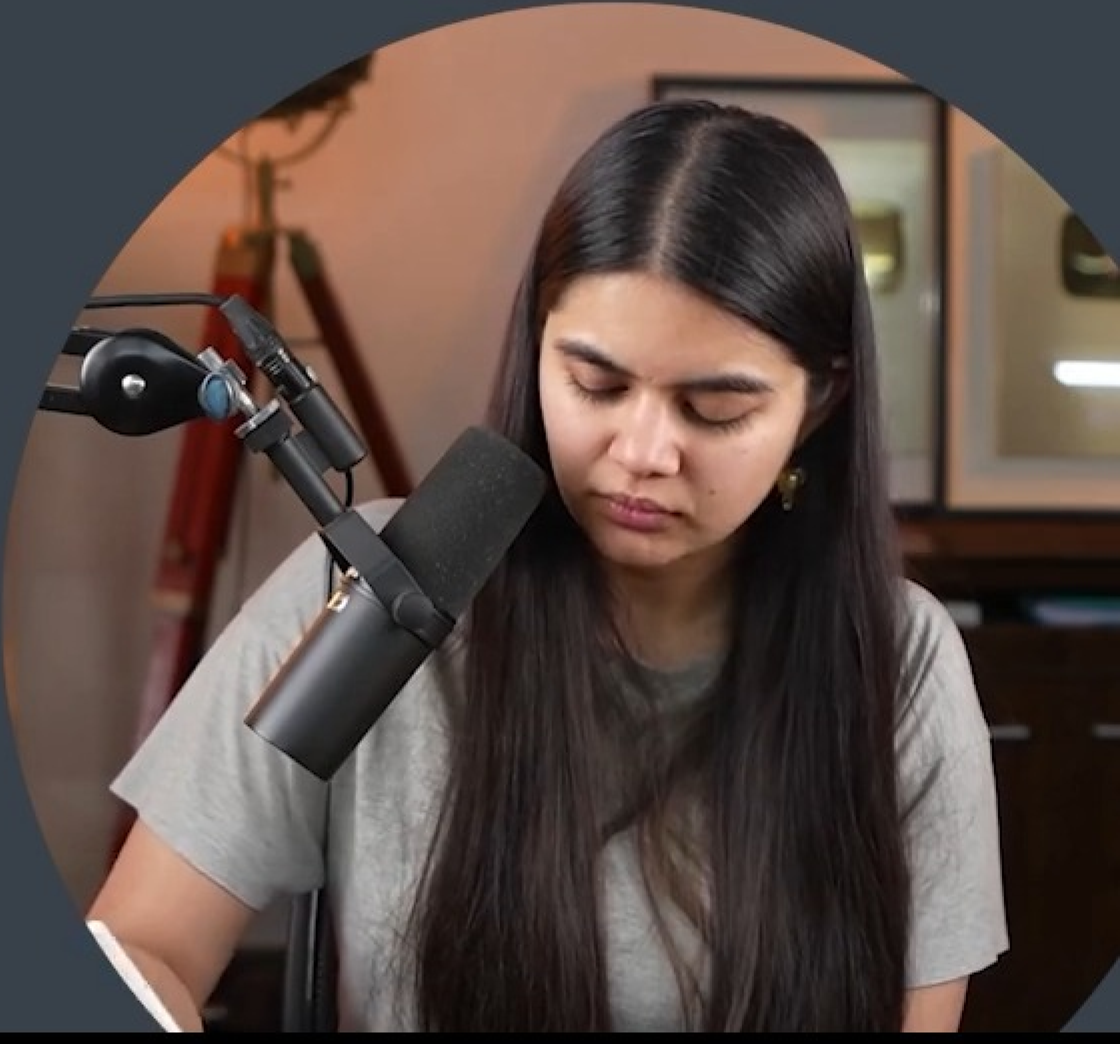
Binary Number System

Decimal to Binary

Convert 37 to binary representation.

(100101)

2	37	1 ✓
2	18	0 ✓
2	9	1 ✓
2	4	0 ✓
2	2	0 ✓
	1 ✓	



$(100\,101)_2 = (37)_{10}$

Binary Number System

Decimal to Binary

Convert 37 to binary representation.

2	37	1 ✓
2	18	0 ✓
2	9	1 ✓
2	4	0 ✓
2	2	0 ✓
	1 ✓	

Binary Number System

Decimal to Binary

Convert 37 to binary representation.

$$1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1$$

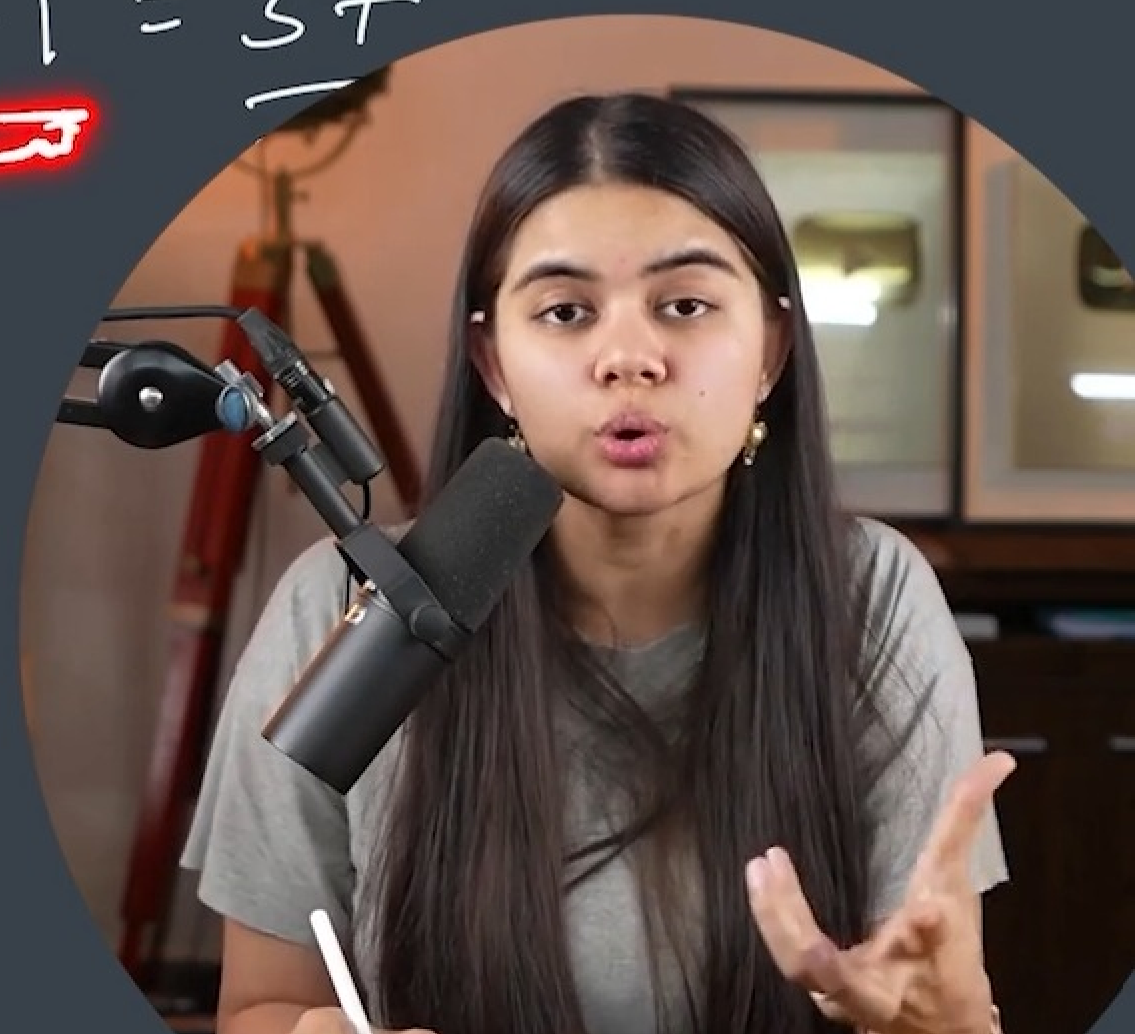
$$2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

$$(100101)_2 = (37)_{10}$$

$$2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

$$1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \qquad 37$$

$$(32) + 4 + 1 = 37$$



Binary Number System

Decimal to Binary

Convert 37 to binary representation.

$<= 25$ 9

2^5	2^4	2^3	2^2	2^1	2^0
0	1	1	0	0	1

$16 + 8$

(11001)

$(100101)_2 = (37)_{10}$

2^6	2^5	2^4	2^3	2^2	2^1	2^0
	1	0	0	1	0	1

37

$(32) + 4 + 1 = 37$

05:22 / 05:41

Decimal to Binary (0 to 16)

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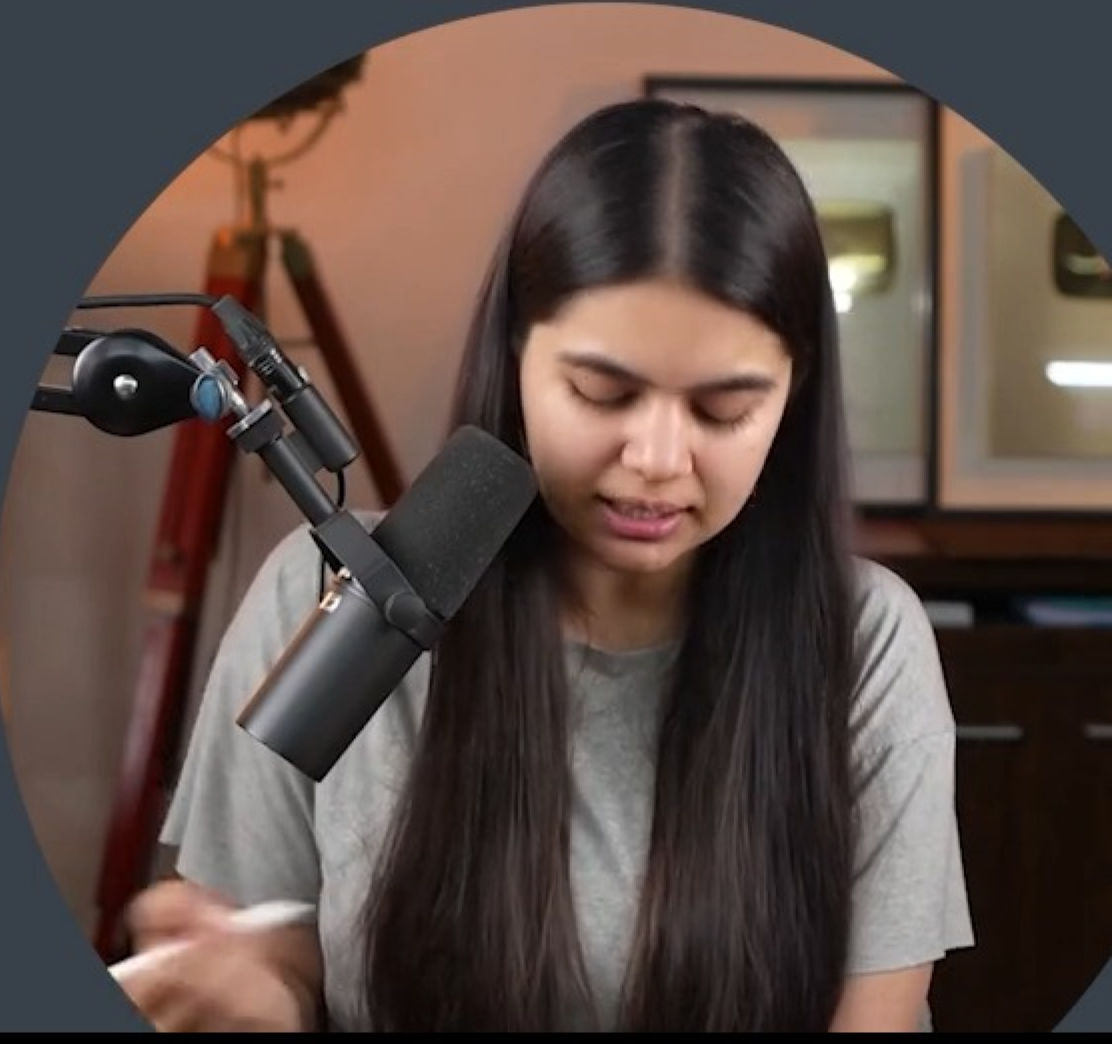
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111
16	10000

$$1 \times 2^0 = 1$$



Decimal to Binary (0 to 16) APNA COLLEGE

1	1
2	1 0
3	1 1
4	1 0 0
5	1 0 1
6	1 1 0
7	1 1 1
8	1 0 0 0
9	1 0 0 1
10	1 0 1 0
11	1 0 1 1
12	1 1 0 0
13	1 1 0 1
14	1 1 1 0
15	1 1 1 1
16	1 0 0 0 0

$2^4 = 16$

$$\begin{array}{r} 1 \\ \times 2^0 \\ \hline 1 \\ \hline \end{array}$$

Data Type **Modifiers**

Alter the meaning of existing data types.

long	>= 4 bytes (more than int)
short	2 bytes
signed	signed int is same as int
unsigned	can only store non-negative numbers
long long	same as long long int

int → -2^{31} to $2^{31}-1$

(4 bytes)

32 bits

0/1 0/1 0/1 _ _ _ _ _ .... _ _ _ _

32 sp

2^{32} ✓

$\frac{2^{32}}{2} = 2^{31}$

Cpp Codes

code.cpp

code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << sizeof(int) << endl;
    cout << sizeof(long int) << endl;
    return 0;
}
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL zsh

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
4
8
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

Ln 6, Col 33 (14 selected) Spaces: 4 UTF-8

Cpp Codes

APNA COLLEGE

code.cpp

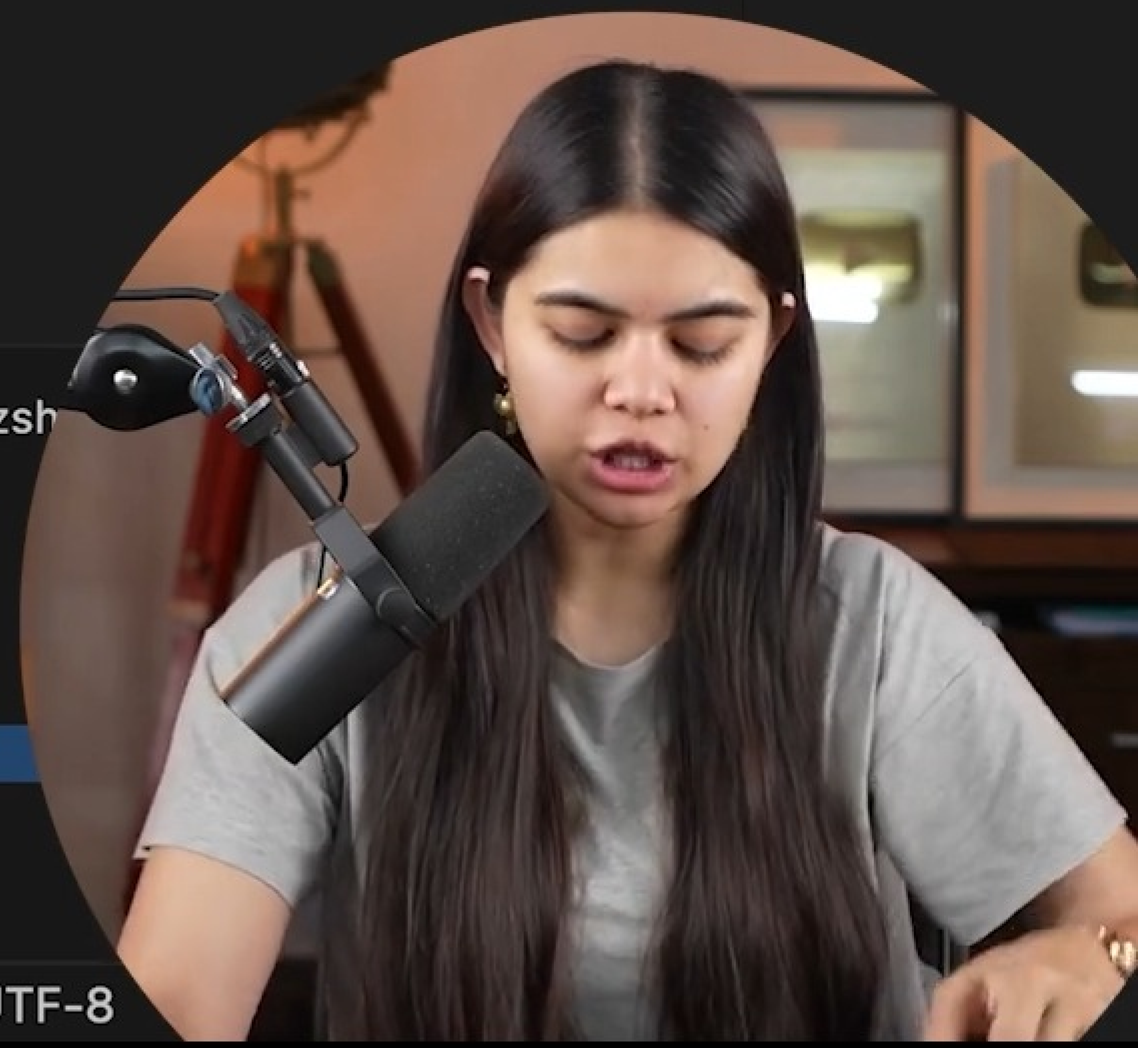
code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << sizeof(double) << endl;
    cout << sizeof(long double) << endl;
    return 0;
}
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL zsh

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
4
8
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
8
8
apnacollege@Amans-MacBook-Pro Cpp Codes %
```



0 0 0 Ln 6, Col 31 Spaces: 4 UTF-8

Cpp Codes

APNA COLLEGE

code.cpp

code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << sizeof(int) << endl;
    cout << sizeof(short int) << endl;
    return 0;
}
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL zsh

```
8
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
8
8
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
4
2
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

0 0 0 Ln 6, Col 25 (5 selected) Spaces: 4 UTF-8

06:30 / 12:13

armankhan082020@gmail.com

Cpp Codes

code.cpp

code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << sizeof(int) << endl;
    cout << sizeof(long long int) << endl;
    return 0;
}
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL zsh

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
4
8
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

⊗ 0 ⚠ 0 0 Ln 6, Col 29 Spaces: 4 UTF-8

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Data Type Modifiers

Alter the meaning of existing data types.

long	>= 4 bytes (more than int)
short	2 bytes
signed	signed int is same as int
unsigned	can only store non-negative numbers
long long	same as long long int

int

↓ 32 bits

0

1

MSB

most significant bit

32 spaces

Data Type **Modifiers**

Alter the meaning of existing data types.

unsigned int → 2^{32} numbers

↓ 32 bits

0

1

MSB

most significant bit

32 spaces

long	>= 4 bytes (more than int)
short	2 bytes
signed	signed int is same as int
unsigned	can only store non-negative numbers
long long	same as long long int

age

+ve

Data Type Modifiers

unsigned short int age ;

Alter the meaning of existing data types.

long	>= 4 bytes (more than int)
short	2 bytes
signed	signed int is same as int
unsigned	can only store non-negative numbers
long long	same as long long int

Convert from Binary to Decimal

n = 101

1 0 1

2^1 2^0

Cpp Codes

APNA COLLEGE

code.cpp

code.cpp > binToDec(int)

```cpp
#include <iostream>
using namespace std;

void binToDec(int binNum) {
    int n = binNum;
    int decNum = 0;
    int pow = 1; //2^0 2^1 2^2 ..

    while(n > 0) {
        int lastDig = n % 10;
        decNum += lastDig * pow;
        pow = pow * 2;
        n = n/10;
    }

    cout << decNum << endl;
}

int main() {
    binToDec(101);
    return 0;
}
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

0 0 0 Ln 18, Col 1 (263 selected) Spaces: 4 UT

03:25 / 04:19

armankhan082020@gmail.com

Cpp Codes

code.cpp ×

code.cpp > main()

```cpp
using namespace std;

void binToDec(int binNum) {
    int n = binNum;
    int decNum = 0;
    int pow = 1; //2^0 2^1 2^2 ..

    while(n > 0) {
        int lastDig = n % 10;
        decNum += lastDig * pow;
        pow = pow * 2;
        n = n/10;
    }

    cout << decNum << endl;
}

int main() {
    binToDec(1010);
    return 0;
}
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
5
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
10
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

Ln 20, Col 18 Spaces: 4 UT

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Convert from Decimal to Binary

n = 7

2	7	1
2	3	1
	1	

/2

1 1 1

$4 \% 2 \Rightarrow 0$

n = 4

2	4	0
2	2	0
	1	

100

$\downarrow$

$1 + 0 + 0$

$10^2 \quad 10^1 \quad 10^0$

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Cpp Codes

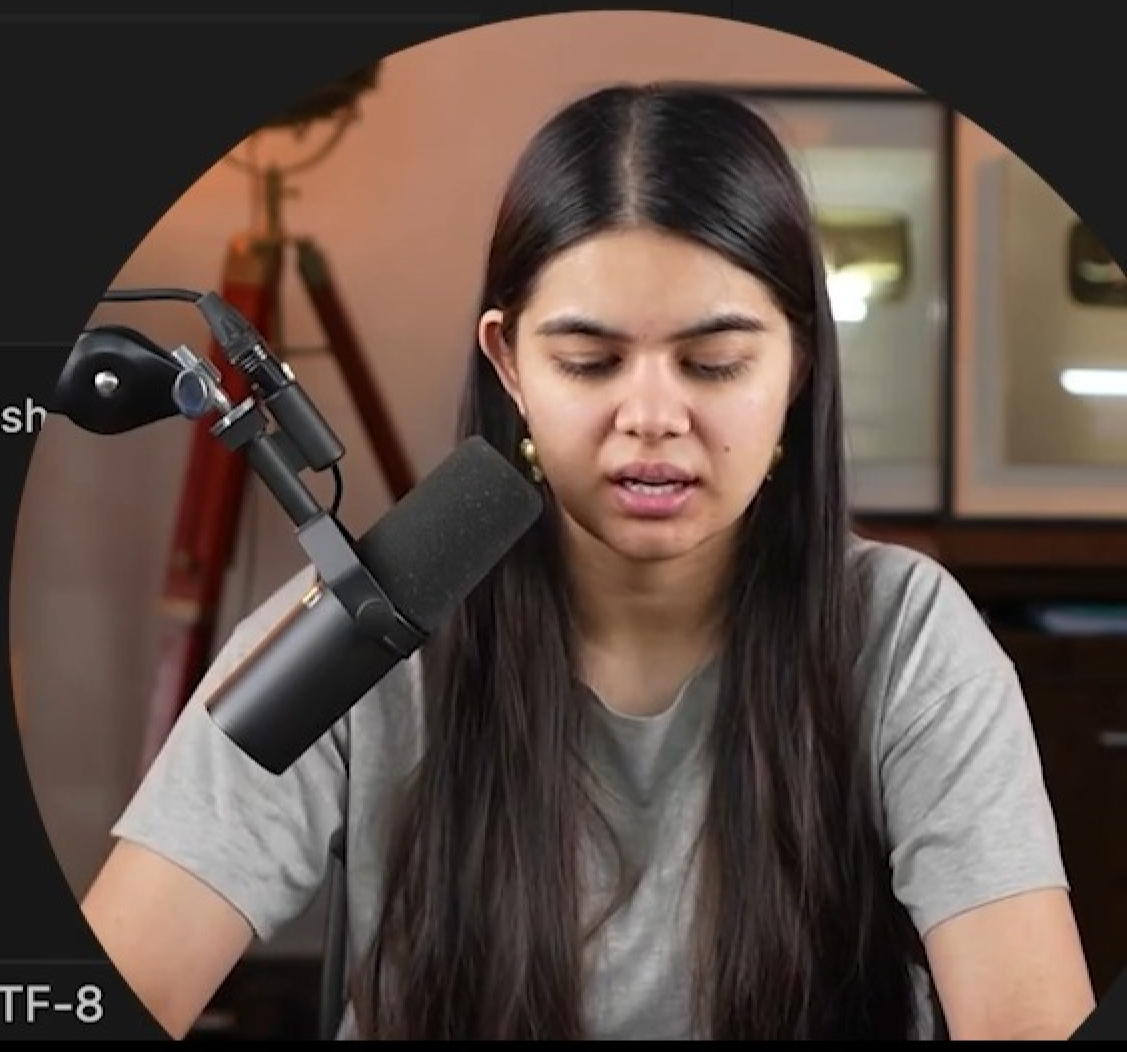
code.cpp

code.cpp > decToBin(int)

```cpp
    int pow = 1; //10^0 10^1 10^2...
    int binNum = 0;

    while(n > 0) {
        int rem = n % 2;
        binNum += rem * pow;
        n = n/2;
        pow = pow * 10;
    }

    cout << binNum << endl;
}

int main() {
    binToDec(1011);
    decToBin(4);
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL zsh

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
5
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
10
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
11
apnacollege@Amans-MacBook-Pro Cpp Codes % cle
```



⊗ 0 ⚠ 0 0 Ln 31, Col 28 Spaces: 4 UTF-8